Supplementary Information

S1. Common-membership AustArch benchmark

S1.1 Source data and filtering

AustArch is a continental radiocarbon and luminescence-age compilation for Australia (Williams et al., 2014; Williams and Ulm, 2014). The source object is the Archaeology Data Service (ADS) collection 1661, DOI `10.5284/1027216`, file `Austarch_1-3_and_IDASQ_28Nov13-1.csv`. A fresh ADS download on 12 August 2026 contained 5,522 rows: 5,044 labelled `Radiocarbon` and 478 assigned to other dating methods. This reconciles the published count of 5,044 radiocarbon determinations with the larger deposited multi-method table.

The analysis used a project working derivative rather than a pristine ADS release. Its pre-analysis edit history is incomplete, but all 976 analysed identifiers occur in the ADS download with matching conventional ages, errors, methods, laboratory codes and analysis-status fields; differences in site, site-type and material labels are confined to spaces replaced by underscores. The source reconciliation, historical differences and checksums are documented in `docs/AUSTARCH_PROVENANCE.md` in the reproducibility archive. The recovered historical workflow also reconstructs the accepted 976-record reference byte for byte from the earliest 5,522-row repository import. This establishes an executable route to the analysed reference, without resolving the rationale for every later change to the larger working file.

Initial screening retained conventional or accelerator-mass-spectrometry radiocarbon determinations with reported ages and one-sigma errors, removed records flagged as erroneous or unrelated to human activity, and restricted conventional ages to 500–20,000 14C yr BP. The historical workflow then restricted material and site types, calibrated terrestrial and marine records with SHCal20 and Marine20, retained determinations whose returned posterior grid overlapped 651–11,650 cal BP, and required at least five dates per site. The resulting filtered reference set contains 976 determinations from 116 sites. Its checksum is verified before it is used by the narrower primary analysis.

The primary empirical display analysis then applied a narrower, prespecified common-membership filter. We retained records classified as terrestrial whose material was `Charcoal`, `Charcoal_from_hearth`, `Scattered_charcoal`, `Wood_charcoal`, `Carbonised_wood`, `Burnt_wood`, `Charred_wood` or `Wood`. Shell determinations were excluded because appropriate local reservoir corrections were unavailable, and human bone was excluded because dietary reservoir effects and data-governance considerations require separate treatment. Records coded `Y` (occupation-related) or `U` (uncertain) were retained. All were calibrated with SHCal20 (Hogg et al., 2020).

For each posterior, we recorded its median, weighted mean and probability mass within 651–11,650 cal BP. A determination was calendar-eligible when its median lay within the one-bandwidth-buffered interval 751–11,550 cal BP and at least 0.80 of its posterior mass lay inside the analysis window. Sites then required at least five eligible dates. The flow was:

Filter stage	Dates	Sites
Filtered reference set	976	116
Terrestrial charcoal or wood	737	106
Common calendar membership	645	106
At least five eligible dates per site	540	68

The final 540 dates comprised 497 `Charcoal`, 28 `Charcoal_from_hearth`, 10 `Scattered_charcoal`, four `Wood_charcoal` and one `Charred_wood` record. Occupation-relation codes were `Y` for 181 dates and `U` for 359. The benchmark is deliberately conditioned on well-replicated sites and is not an unbiased sample of Sahul occupation.

S1.2 Curve construction

Every curve method received exactly the same selected determinations on the same 100-year bins. The summed probability distribution (SPD) used posterior probability mass within the analysis window, followed by a centred 100-year running mean with unsmoothed edge cells. The frequency distribution (FD) assigned each posterior median to a 100-year bin and applied a centred two-bin running mean for display. The kernel density estimate (KDE) placed Gaussian kernels of 100-year bandwidth on the same posterior medians and reflected them at 651 and 11,650 cal BP. Each resulting curve was normalised to unit integral. Thus, the estimand is relative temporal shape, not total date abundance.

The all-date pairwise total-variation distances were 0.0459 for SPD–KDE, 0.0884 for FD–KDE and 0.1015 for SPD–FD. These values compare construction methods on identical dates; they are not sampling-error estimates.

S1.3 Retention rules and realisations

We compared all dates with random-1, random-5, oldest, youngest and bracketed (oldest plus youngest) retention. A rule treated either 34 of 68 sites (50%) or all 68 sites (100%); untreated sites retained every eligible date. Treated-site subsets were paired across rules. Random retention and half-site treatment used 1,000 realisations from master seed 1234. Full-site oldest, youngest and bracketed retention were deterministic.

Oldest and youngest refer to the greatest and least calibrated posterior median within a site. Median ties were resolved by the posterior weighted mean in the same direction and then by ascending AustArch record identifier. No site had an oldest-median tie. Two sites had youngest ties at both the median and mean stages, so the stable identifier supplied the final decision. These extrema were defined after calibration and should not be interpreted as site onsets or terminations.

For each method and realisation, we calculated total variation from its all-date baseline,

$$D_{\mathrm{TV}}(p, q) = \frac{1}{2} \sum_{t} |p_t - q_t|,$$

where p_t and q_t are normalised bin probability masses. We also calculated all three between-method distances on the retained dates. The conservative ordering contrast was

$$C = \min_{m} D_{\mathrm{TV}}(p_{m,\mathrm{retained}}, p_{m,\mathrm{all}}) - \max_{m \neq m'} D_{\mathrm{TV}}(p_{m,\mathrm{retained}}, p_{m',\mathrm{retained}}).$$

A deterministic scenario passed the retention-versus-method gate when $C > 0$; a stochastic scenario required the 2.5th percentile of C to exceed zero. Temporal displacement was the change in mean cal BP, using bin midpoints. An older result required positive shifts for all methods (positive 2.5th percentiles when stochastic); a younger result required negative shifts (negative 97.5th percentiles).

The following full-site SPD comparisons show retention effects at a fixed retained count for oldest, youngest and random-1. Random-rule entries are means followed by central 95% ranges across 1,000 conditional retention realisations.

Rule	Dates	SPD D_{TV}	SPD mean-cal-BP shift (yr)
Oldest	68	0.455	+3,133
Youngest	68	0.524	−2,439
Random-1	68	0.204 (0.158–0.256)	+103 (−433 to +594)
Bracketed	136	0.279	+352
Random-5	340	0.065 (0.052–0.080)	+95 (−37 to +227)

These are descriptive comparisons of the accepted outputs, not a new hypothesis test. The bracketed mean shift falls within the random-1 shift range, but its SPD distance exceeds the 97.5th percentile of random-1 distances. Moreover, random-1 retains half as many dates as bracketed retention; no empirical random-2 comparison was made.

The prespecified retention-versus-method criterion is a separate comparison. Only all-site oldest, youngest and bracketed retention met it, with contrasts of +0.193, +0.370 and +0.084, respectively. Their method-specific distances were 0.455–0.535, 0.524–0.554 and 0.279–0.326; the largest between-method distances were 0.263, 0.154 and 0.195. Neither half-site extremum, either random rule at either proportion, nor half-site bracketed retention met the criterion. Figure 6 envelopes describe conditional retention realisations; they do not estimate uncertainty over sites, archive inclusion or population history.

S1.4 Frozen sensitivity suite

The nine configurations were prespecified for oldest and youngest retention. Before the bracketed extension was run, the same configurations and all-nine decision gate were frozen for that rule. The suite comprised calendar buffers of 0, 100 and 200 years; posterior-mass thresholds of 0.50, 0.80 and 0.95;

reflected KDE bandwidths of 50, 100 and 200 years; an unreflected 100-year KDE; and the occupation-related `Y`-only subset. The configurations retained 496–576 dates from 62–73 sites, except the `Y`-only set, which retained 154 dates from 24 sites. Every configuration preserved a positive ordering contrast and older shift for oldest retention, and a positive contrast and younger shift for youngest retention. Bracketed retention shifted older in all nine configurations and had a positive ordering contrast in eight, ranging from +0.060 to +0.089; the `Y`-only contrast was -0.054. Oldest and youngest are therefore sensitivity-robust within this suite, whereas bracketed retention is reported as a primary-specification result.

S2. Simulation of constant and changing histories

S2.1 Empirical site-count and reported-error models

Simulation parameterisation used the 976-date, 116-site reference set rather than the narrower display benchmark. The empirical number of dates per site had mean 8.414, median 7 and range 5–33. A negative-binomial distribution left-truncated at five dates was fitted by maximum likelihood. After conditioning at five, the fitted mean was 8.412 (Supplementary Fig. S2). The untruncated parameter estimates lie near the boundary: their conditional shape approaches a truncated logarithmic-series distribution with mass proportional to q^k/k for $k \geq 5$, where $q = 0.8489$. This is a well-defined conditional count model used to reproduce the observed count shape; we attach no interpretation to the untruncated parameters, which are retained in the code and provenance records. Each synthetic replicate drew 116 site counts from this distribution.

Reported uncertainty was modelled as

$$\log(\sigma_{\text{quoted}}) = 4.0992 + 8.5643 \times 10^{-5}\,\text{CRA} + \varepsilon,$$

with residual standard deviation 0.4719 and $R^2 = 0.170$. This regression reproduces the broad age–error relation; it is not a laboratory-performance model.

S2.2 Generating truths and observation model

Five unit-integral regional event-age densities were defined over 651–11,650 cal BP: uniform, linear increase, linear decrease, exponential increase and exponential decrease (Supplementary Fig. S1). "Increase" means greater density at the recent, low-cal-BP boundary. In the main experiment, every non-uniform truth had a ten-fold endpoint ratio, summarised on the common scale

$$\theta = \log\left\{\frac{f(651)}{f(11650)}\right\}.$$

Thus $\theta = \log(10)$ for ten-fold increasing truths, $-\log(10)$ for decreasing truths and 0 for uniform truth. Linear slopes and exponential rates were signed, so both fitted families could represent either temporal direction.

For each replicate, all calendar ages were independent draws from the regional truth. Site labels determined only how many dates were grouped together and which observations were subsequently retained. They did not impose occupation windows, durations, stratigraphic order or within-site dependence.

The primary observation model mapped each calendar age to the central conventional radiocarbon age expected from SHCal20 (`ccCRA` from `rcarbon::uncalibrate()`), drew a quoted one-sigma error from the empirical regression and recalibrated the central age. It therefore propagated posterior width and calibration-curve structure but conditioned on the calibration-curve-mean conventional age.

S2.3 Retention design and stochastic control

Eight rules were evaluated: all dates; random-1, random-2, random-3 and random-5 without replacement; oldest and youngest by calibrated posterior median; and bracketed retention of both extrema. Each rule treated 50%, 75% or 100% of the 116 sites, with untreated sites retaining all dates. All counts were at least five, so every random rule was feasible. We generated 1,000 replicates for each of five truths at each of three proportions. Replicate i used deterministic seed $1234 + i$ under `L'Ecuyer-CMRG`, with common random numbers preserving paired comparisons. Random-2 was drawn on an isolated RNG state so that adding it could not change dates, errors, treated sites or any previously implemented rule.

The accepted matrix comprises 15 scenario files, 1,000 replicates in each truth-by-proportion cell and 720,000 fitted-result rows. Replicates are paired across proportions: the all-dates result for a truth is exactly identical across the three treatment files. The 15,000 scenario-replicate combinations should therefore not be interpreted as independent simulated datasets. At full treatment, random-1, oldest and youngest each retain 116 dates; random-2 and bracketed retain 232, random-3 retains 348 and random-5 retains 580. Untreated sites retain all their dates, so a site-treatment proportion is not a retained-date proportion.

S2.4 Model fitting and selection

The fitted regional density $f(x \mid \theta)$ was evaluated against each calibrated posterior $p_i(x)$. The primary marginal log likelihood was

$$\ell_i^1 = \log \int f(x \mid \theta) p_i(x)\, dx.$$

The secondary score was

$$\ell_i^2 = \int \log\{f(x \mid \theta)\} p_i(x)\, dx.$$

Calibration used a uniform calendar-age prior. Each calibrated posterior was restricted to the strict interior of the 651–11,650 cal BP analysis window and renormalised before fitting. Conditional on posterior mass within this window, the first criterion is the observed-data marginal log likelihood up to a date-specific additive constant independent of θ; these constants cancel in within-dataset AIC comparisons (Akaike, 1974; Timpson et al., 2021). The second is the posterior expected log density; we applied the same parameter penalty as a scoring sensitivity but do not describe it as a marginal likelihood or its criterion as AIC. Integrals were finite sums on the calibrated calendar grids.

A uniform candidate and linear and exponential candidates of either direction were fitted by bounded numerical optimisation. With $L = 11650 - 651 = 10999$ years, the linear slope a lies in $[-2/L^2, 2/L^2]$, its natural nonnegative-density domain. The exponential rate r is restricted to $[-0.001, 0.001]$ yr^{-1}, an imposed finite cap. Optimisation used tolerance 10^{-6} on the scaled parameters $10^6 a$ and $10^3 r$. The original near-boundary flag means that the scaled estimate is within 5×10^{-6} of either bound; it does not certify an exact endpoint optimum.

Minimum AIC under ℓ^1 supplied the primary selected family. The criterion assumes independent regional draws and does not include retention probabilities. For bracketed retention, the oldest and youngest observations retained from a site are dependent; the product criterion treats their posteriors under working independence. Bracketed AIC and family-selection results therefore describe this analysis workflow rather than likelihood support under a joint paired-order-statistic model. We recorded the selected direction state (uniform, increasing or decreasing), generating-family selection, direction reversal, fitted generating-family θ and boundary frequency. Truth-consistent state means uniform selection under uniform truth and the correct direction under a non-uniform truth, irrespective of functional family.

Figure 4 summarises the bounded endpoint contrast $c = \tanh(\theta/2)$, which equals $aL^2/2$ for linear fits and $\tanh(rL/2)$ for exponential fits. At the natural linear endpoints, $c = \pm 1$ while $\theta = \pm\infty$; at the exponential cap, $|\theta| = 10.999$ and $|c| = \tanh(10.999/2) < 1$. The figure transforms each stored generating-family fit regardless of which family AIC selected, then calculates the mean and type-7 2.5th and 97.5th percentiles across all 1,000 replicates. These are empirical replicate ranges, not confidence intervals for a historical parameter. Fractions beside every group retain the original near-boundary flags. Even random-1 produced near-boundary linear fits in 7.2% and 8.2% of full-treatment replicates under increasing and decreasing truths, respectively, so finite log-ratio means at the linear boundary would depend on optimiser tolerance. This derived presentation preserves the original fits and model-selection results without reclassifying any fit as an exact boundary optimum.

S2.5 Direction and family selection

The full-site ten-fold results below report "family; state" percentages under the primary marginal likelihood. For uniform truth, state is uniform; for non-uniform truth, state is the correct direction. An adverse one-sided rule is oldest for increasing truths and youngest for decreasing truths.

Truth	All dates	Oldest	Youngest	Bracketed
Uniform	84.8; 84.8	0.0; 0.0	0.0; 0.0	99.9; 99.9
Linear increase	97.6; 100	19.5; 0.0	0.0; 100	0.0; 100
Exponential increase	99.6; 100	41.6; 0.0	100; 100	100; 100
Linear decrease	97.4; 100	0.0; 100	16.8; 0.0	0.0; 100
Exponential decrease	99.8; 100	100; 100	43.3; 0.0	100; 100

Full oldest retention selected a decreasing direction and full youngest retention an increasing direction in every replicate under every truth. When the rule opposed a non-uniform truth, the directional-reversal rate was 1.000. Under uniform truth, oldest and youngest selected the corresponding exponential direction in all replicates. Concordant one-sided retention frequently drove fitted effects to their bounds. Under the working-independence criterion, bracketed retention preserved direction but selected exponential in every full-treatment linear-truth replicate.

For ten-fold changes, the full-treatment reversal did not extend to partial treatment. At proportions 0.50 and 0.75, no ten-fold non-uniform replicate selected the opposite direction under the adverse rule; at 0.75, uniform was instead selected in 18.1% of linear-increase and 21.2% of linear-decrease replicates. Uniform truth was affected sooner: half-site oldest and youngest retention selected a directional model in 97.3% and 97.9% of replicates. Weaker changes were reversed at partial treatment (Section S2.7).

Across random-1, random-2, random-3 and random-5, all four ten-fold non-uniform truths and all three treatment proportions, direction selection was 1.000. Generating-family selection ranged from 0.754 to 0.995 as information was discarded. The ℓ^1 and ℓ^2 analyses agreed on the selected family-and-direction combination in 99.75% of 120,000 paired selections, including all full-treatment one-sided cells.

S2.6 Order statistics and the limits of model selection

For an idealised site containing n independent, exactly observed calendar ages with density $f(x)$ and cumulative distribution $F(x)$, the maximum (oldest cal-BP age) has cumulative distribution $F(x)^n$, because every age must be at most x. The minimum has cumulative distribution $1-\{1-F(x)\}^n$. Differentiating gives

$$g_{\text{old}}(x) = nF(x)^{n-1}f(x), \qquad g_{\text{young}}(x) = n\{1-F(x)\}^{n-1}f(x).$$

The pooled marginal density of bracketed dates is $\{g_{\text{old}}(x) + g_{\text{young}}(x)\}/2$. For uniform ages on $[a, b]$, writing $u = (x-a)/(b-a)$ gives

$$g_{\text{bracket}}(x) = \frac{n}{2(b-a)}\left[u^{n-1} + (1-u)^{n-1}\right].$$

This density is exactly uniform for $n = 2$, when both observations are retained, and symmetric and U-shaped for every $n \geq 3$. All site counts used here satisfy $n \geq 5$, and mixing over these counts preserves symmetry and the U-shape. The fitted uniform, linear and exponential families cannot represent this shape. Thus, selecting uniform under bracketed uniform truth need not indicate that the retained age distribution is uniform: it may indicate only that neither monotonic alternative fits better after the parameter penalty. Similarly, generating-family selection under non-uniform truth is a comparison within the candidate set, not an absolute goodness-of-fit test. These formulae describe exact-age selection. The simulation instead selects extrema by calibrated posterior median and fits their full posteriors; the formulae explain the mechanism without replacing that observation model or its empirical results.

S2.7 Signal-strength extension

The extension was prespecified on 18 October 2026, before any of its results existed (`docs/STAGE6_EFFECT_SIZE.md` in the archive). It asked how reversal, direction recovery and uniform selection depend jointly on the true end-to-end ratio and the proportion of sites treated. The four changing truths were repeated at ratios of 1.5, 2, 3 and 5 at each treatment proportion, giving 48 scenarios. The simulation code, input data, container image and seeds were those of the main experiment, verified against all seven recorded source and environment fingerprints. Replicate i again used seed $1234 + i$, independent of the ratio, so latent draws were paired across ratios. The planned 1,000 replicates per scenario were reduced to the first 500 before any rate was computed, halving the runtime; the largest possible Monte Carlo standard error for

a rate is therefore 0.022. Ratio-10 and constant-history (ratio 1) values are the accepted 1,000-replicate main-experiment results.

Technical acceptance was checked before interpretation. A 20-replicate rerun of the accepted linear-increase, full-treatment, ten-fold scenario reproduced the first 20 accepted replicates exactly in all 17 result columns. Every scenario file was complete, matched its expected truth, proportion, ratio and replicate count, and carried the accepted provenance. Every truth integrated to one with the requested endpoint ratio. A separate aggregation of all 1,872,000 stored fitted rows, written without the production summary functions, reproduced every reported rate. The ℓ^1 and ℓ^2 analyses selected the same family and direction in 99.69% of 312,000 paired selections.

Main Figure 3 plots both the correct-state and the reversal rate. The prespecified display showed only the correct-state rate, which cannot distinguish reversal from uniform selection. The reversal rate, also a prespecified outcome, was therefore added to the display after the results were seen; the prespecified display is retained in the archive. For changing histories, the uniform-selection rate is 100% minus the correct and reversed percentages.

Under the opposing one-sided rule (oldest for increasing truths, youngest for decreasing truths), each cell below gives correct direction; reversed direction, as percentages of replicates under the primary marginal likelihood. Columns are true end-to-end ratios. Ratios 1.5–5 have 500 replicates and ratio 10 has 1,000.

Truth	Sites treated	1.5	2	3	5	10
Linear increase	50%	1.8; 19.6	37.6; 0.0	98.6; 0.0	100; 0.0	100; 0.0
Linear increase	75%	0.0; 99.8	0.0; 94.4	0.6; 44.2	17.8; 2.8	81.9; 0.0
Linear increase	100%	0.0; 100	0.0; 100	0.0; 100	0.0; 100	0.0; 100
Linear decrease	50%	0.2; 22.6	33.6; 0.4	97.6; 0.0	100; 0.0	100; 0.0
Linear decrease	75%	0.0; 100	0.0; 98.6	0.0; 50.0	14.0; 2.4	78.8; 0.0
Linear decrease	100%	0.0; 100	0.0; 100	0.0; 100	0.0; 100	0.0; 100
Exponential increase	50%	2.0; 19.0	43.2; 0.0	100; 0.0	100; 0.0	100; 0.0
Exponential increase	75%	0.0; 99.8	0.0; 93.4	1.0; 35.2	49.8; 0.4	100; 0.0
Exponential increase	100%	0.0; 100	0.0; 100	0.0; 100	0.0; 100	0.0; 100
Exponential decrease	50%	0.2; 22.2	40.6; 0.2	99.4; 0.0	100; 0.0	100; 0.0
Exponential decrease	75%	0.0; 100	0.0; 97.8	0.0; 39.4	45.0; 0.2	100; 0.0
Exponential decrease	100%	0.0; 100	0.0; 100	0.0; 100	0.0; 100	0.0; 100

The next table gives correct-direction percentages for the other rules, as the range across the four changing truths. The ratio-1 column gives uniform selection under the constant history. No random rule reversed more than 0.6% of replicates in any changing-history scenario, and bracketed retention and all dates reversed none.

Rule	Sites treated	1	1.5	2	3	5	10
Bracketed	50%	88.4	89.6–92.2	99.8–100	100	100	100
Bracketed	75%	91.0	70.6–78.6	98.6–99.6	100	100	100
Bracketed	100%	99.9	10.6–16.8	69.6–79.6	99.8–100	100	100
Random-1	50%	84.6	89.2–92.8	99.8–100	100	100	100
Random-1	75%	82.9	72.6–78.8	97.0–98.8	100	100	100
Random-1	100%	82.4	40.4–49.2	71.2–79.4	96.0–97.6	99.4–100	100
Random-2	50%	84.1	92.2–94.4	99.8–100	100	100	100
Random-2	75%	83.7	82.4–85.4	99.4–99.6	100	100	100
Random-2	100%	83.1	60.6–66.6	93.2–95.2	99.8–100	100	100
All dates	all	84.8	98.2–99.6	100	100	100	100

Bracketed retention avoided reversal but often selected uniform for weak changes, more often than its count-matched random-2 comparison. It also never selected the linear family for a linear history at 75% or 100% treatment at any ratio, and did so for at most 3.5% at 50%. Random-2 selected the linear family for 32.0–94.6% of linear histories. Under the concordant one-sided rule, every changing history at every ratio and treatment proportion was given the correct direction. The complete rates for every rule, truth, ratio, proportion and score, including generating-family selection and ℓ^1/ℓ^2 agreement, are in `out/revision/effect-size-20261018/stage6_selection_rates.csv`.

S2.8 Realised measurement scatter

The rerun was prespecified on 23 October 2026, after the earlier assessment in Section S3 (`docs/STAGE7_REALISED_NOISE.md` in the archive). It repeated the main ten-fold matrix of five truths, three treatment proportions, eight rules and 1,000 replicates, with every conventional age drawn from a normal distribution centred on the curve-mean age, with standard deviation $\sqrt{\sigma^2_{\text{curve}} + \sigma^2_{\text{quoted}}}$, rounded and recalibrated with its quoted error. The draws used an isolated random-number stream. Calendar ages, quoted errors, sites, treated sites and random selections were therefore identical to the main experiment; only conventional ages, posteriors and posterior-median extrema differed. As in Section S3, a retained date with no posterior mass inside the strict window was dropped and counted; this affected at most 0.011% of retained dates. That rule was adopted after the first launch stopped on such a date and before any rate was computed.

Before interpretation, the rerun code reproduced 20 accepted replicates exactly under the curve-mean model, six paired datasets confirmed identical structural draws, and all 15 scenarios were complete with the accepted provenance. A separate aggregation from the 1,440,000 stored fitted rows, written without the project's summary functions, reproduced every rate.

Headline result	Rates	Curve mean	Realised	Largest change
Full opposing-rule reversal	4	1.000	1.000	0.000
Full concordant-rule direction	4	1.000	1.000	0.000
Constant, full oldest/youngest: uniform	2	0.000	0.000	0.000

Headline result	Rates	Curve mean	Realised	Largest change
Constant, half oldest/youngest: directional	2	0.973–0.979	0.972–0.980	0.001
Partial opposing-rule reversal	8	0.000	0.000	0.000
Random-rule reversal	48	0.000	0.000	0.000
Full bracketed: constant uniform	1	0.999	0.999	0.000
Full bracketed: direction	4	1.000	1.000	0.000
Full bracketed: linear family	2	0.000	0.000	0.000

Across all cells, the largest change in any rate was 0.015, in a generating-family rate. Each cell had at least 95.2% of replicates selecting the same family and direction under both observation models; the median was 99.1%. Scatter changed which date was a site's posterior-median oldest for about 0.2–1.5% of dates in the paired check. The rerun covers the ten-fold design only, with independent curve errors and no laboratory offsets or outliers.

S3. Earlier realised-observation-noise sensitivity

The current analysis retained, but did not extend, an earlier paired sensitivity based on uniform, linear-increase and exponential-increase truths. Its candidate set comprised uniform and growth-only linear and exponential models; it did not include the decreasing linear and exponential candidates used in the current signed competition. Historical "accuracy" therefore means selection of the generating family within that restricted candidate set. The sensitivity covered all dates, random-1, random-3 and random-5 at the three treatment proportions, with 1,000 pairs per truth-by-proportion cell. Each pair shared the latent ages, site treatment, retained observations and structural random-number state. In the realised arm,

$$r_{\mathrm{CRA}} = \mathrm{round}\left[ccCRA + Z\sqrt{ccError^2 + \sigma^2_{\mathrm{quoted}}}\right], \qquad Z \sim N(0,1),$$

where `ccError` is SHCal20 curve error and σ_{quoted} is the simulated laboratory error. Realised ages were recalibrated. Observations with zero posterior mass inside the strict likelihood window were dropped and recorded. Oldest, youngest and bracketed retention were not included.

This earlier analysis also recorded unique generating-family selection under a prespecified parsimony rule. Models whose total criterion difference from the minimum did not exceed the estimated standard error of their pointwise differences were treated as similarly supported, after which only the lowest-dimensional supported model or models were retained. A replicate counted as unique support only when the generating family alone remained. This outcome is retained for audit continuity but is not used for the main results.

The prespecified v1 screen used 5,000 paired cluster-bootstrap resamples and simultaneous 95% max-deviation intervals over marginal arms and paired differences. It met 15 of 18 prespecified criteria and therefore is not treated as passed. Observed accuracy differences ranged from -0.015 to +0.008 and unique-support differences from -0.019 to +0.009. The pooled v1 maximum absolute endpoints were 0.0373 for accuracy and 0.049 for support against 0.03 margins; truth-specific endpoints were 0.047 and 0.063 against 0.05 margins. None of 72 realised-minus-mean boundary-rate differences was positive, but the symmetric simultaneous upper endpoint was 0.019 against a threshold below 0.01.

After that assessment was inspected, a post-unblinding diagnostic (labelled v2 in the archive) recalibrated intervals over only the paired-difference estimands implicated by those failures. Its pooled/truth-specific endpoints were 0.0203/0.030 for accuracy, 0.024/0.038 for unique support and 0.008/0.008 for the one-sided boundary metric, within the respective 0.03, 0.05 and 0.01 margins. Two independent 50,000-resample seeds gave identical critical widths, endpoints and decisions. The diagnostic supports describing

the tested point shifts as small. It does not replace the failed prespecified assessment. Section S2.8 tests oldest, youngest and bracketed retention separately.

S4. Reproducibility and archive status

The accepted empirical and simulation evidence was generated under R 4.5.3 in a recorded container image. Scenario metadata, deterministic seeds and checksums connect the results to their generation sources. The empirical records document filters, effective counts, curve distances, tie resolution and sensitivity decisions. Scientific tests protect density calculations, retention, stochastic pairing and manuscript-facing transformations; separate checksum checks verify artefact integrity. The archive README gives the commands for these checks and for resumable reruns.

The supplementary reproducibility archive, supplied in anonymised form for peer review, contains the active code, pinned environment, focused tests, synthetic simulation outputs, aggregate empirical evidence and provenance snapshots. It excludes raw and record-level AustArch data, including source site identities, coordinates, laboratory identifiers and human-remains records. A citable repository record containing code, synthetic artefacts and permitted aggregate evidence will be made public no later than publication. Any release of source or record-level archaeological data remains conditional on licensing, sensitive-location review and Indigenous data-governance requirements.

References

Akaike, H., 1974. A new look at the statistical model identification. IEEE Transactions on Automatic Control 19, 716–723. https://doi.org/10.1109/tac.1974.1100705

Hogg, A.G., Heaton, T.J., Hua, Q., Palmer, J.G., Turney, C.S., Southon, J., Bayliss, A., Blackwell, P.G., Boswijk, G., Ramsey, C.B., Pearson, C., Petchey, F., Reimer, P., Reimer, R., Wacker, L., 2020. SHCal20 Southern Hemisphere calibration, 0-55,000 years cal BP. Radiocarbon 62, 759–778. https://doi.org/10.1017/RDC.2020.59

Timpson, A., Barberena, R., Thomas, M.G., Méndez, C., Manning, K., 2021. Directly modelling population dynamics in the South American Arid Diagonal using 14C dates. Phil. Trans. R. Soc. B 376, 20190723. https://doi.org/10.1098/rstb.2019.0723

Williams, A., Ulm, S., 2014. AustArch: A database of 14C and luminescence ages from archaeological sites in Australia. https://doi.org/10.5284/1027216

Williams, A., Ulm, S., Smith, M., Reid, J., 2014. AustArch: A database of 14C and Non-14C Ages from archaeological sites in Australia - composition, compilation and review. IA. https://doi.org/10.11141/ia.36.6

Supplementary Information

Fig. S1.

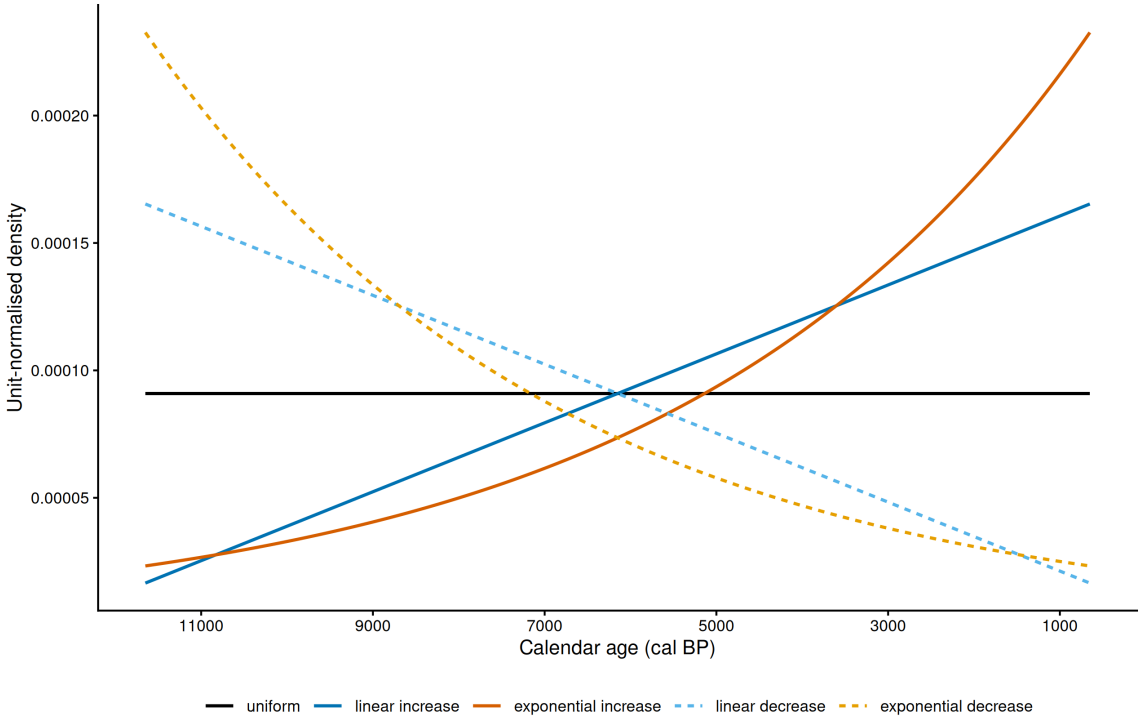


Figure 1: The five unit-normalised regional event-age densities used in the simulation. Increasing trajectories have greater density at the recent (low-cal-BP) boundary; decreasing trajectories have greater density at the old boundary. Each non-uniform trajectory has the main experiment's ten-fold end-to-end ratio, $|\theta| = \log(10)$, over 651–11,650 cal BP. Colour and line type jointly identify the truth.

Fig. S2.

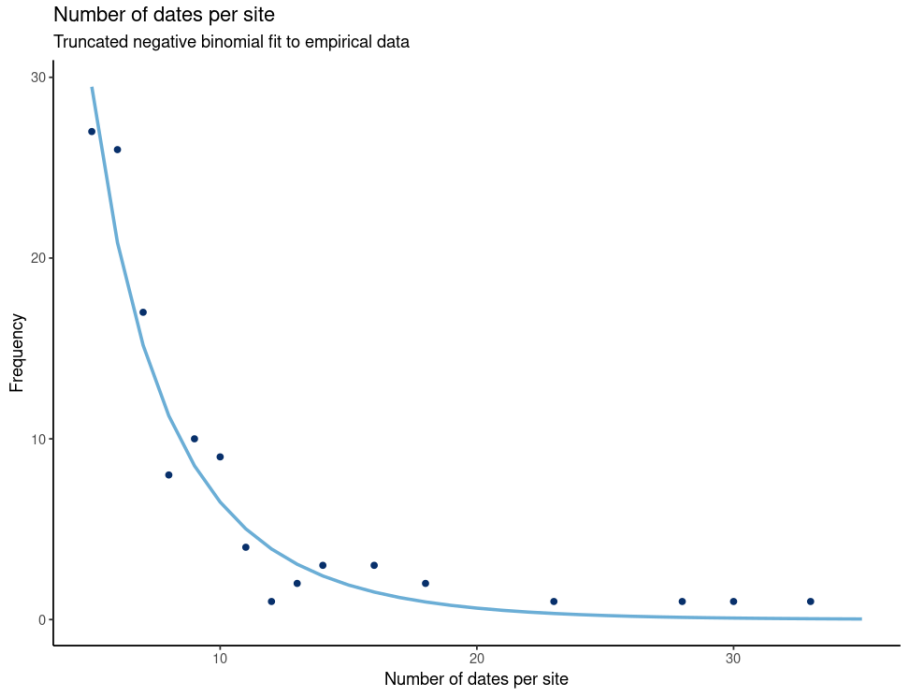


Figure 2: Empirical and fitted distribution of radiocarbon determinations per site in the 976-date, 116-site reference set. Points show the observed frequency; the line is the fitted negative-binomial distribution left-truncated at five dates. The empirical and fitted conditional means are 8.414 and 8.412 dates per site, respectively; the empirical range is 5–33.

Fig. S3.

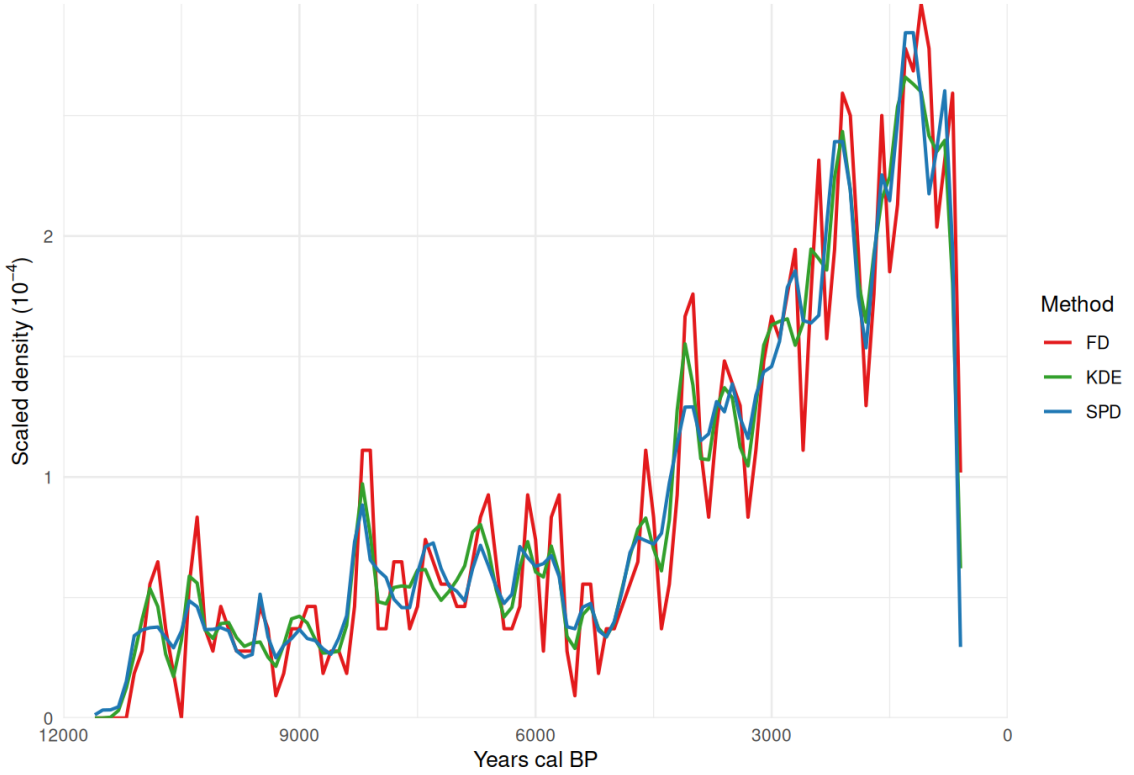


Figure 3: All-date calendar-age summaries for the common-membership empirical benchmark: 540 terrestrial charcoal or wood determinations from 68 AustArch sites. The SPD sums calibrated posterior mass and applies a centred 100-year running mean; the FD bins posterior medians at 100-year resolution and applies a two-bin display smooth; the reflected Gaussian KDE uses the same medians and a 100-year bandwidth. Every method receives the same dates and is normalised to unit integral over 651–11,650 cal BP. Pairwise total-variation distances are 0.046 (SPD–KDE), 0.088 (FD–KDE) and 0.101 (SPD–FD). The padded plotting range is 12,000–0 cal BP; the curves describe relative temporal shape, not date count or population abundance.